

Joris Koefler

RESEARCH INTEREST

Positive Geometry and its applications in QFTs

EDUCATION

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|-----------------------|---|
| 2023 - present | MPI for Mathematics in the Sciences , Leipzig, GER <ul style="list-style-type: none">• PhD in Mathematics<ul style="list-style-type: none">◦ Non-linear algebra group◦ Advisor: Rainer Sinn |
| 2022 - 2023 | University of Warwick , Coventry, UK <ul style="list-style-type: none">• M.Sc. in Mathematics (1-year program) - First Class<ul style="list-style-type: none">◦ Advisor: Diane Maclagan◦ Thesis: A computational approach for tropical solutions to enumerative problems |
| 2020 - 2022 | Ruhr University Bochum , Bochum, GER <ul style="list-style-type: none">• B.Sc in Mathematics (3-year program) - Excellency Award<ul style="list-style-type: none">◦ Advisor: Christian Stump◦ Thesis: Combinatorial models for the positive Grassmannian with a view towards Amplituhedra |
| 2018 - 2020 | Ruhr University Bochum , Bochum, GER <ul style="list-style-type: none">• B.Sc. in Physics (3-year program) - First Class<ul style="list-style-type: none">◦ Advisor: Evgeny Epelbaum◦ Thesis: Application of a subtractive renormalisation scheme in Nucleon-Nucleon scattering in Effective Field Theory |
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TEACHING

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| 2024 - 2024 | Leipzig University , Leipzig, GER <ul style="list-style-type: none">• Teaching Assistant<ul style="list-style-type: none">◦ Combinatorial Commutative Algebra |
| 2019 - 2022 | Ruhr University Bochum , Bochum, GER <ul style="list-style-type: none">• Teaching Assistant<ul style="list-style-type: none">◦ Mathematical Methods for Physicists I + II◦ Classical Theoretical Physics |

2019 - 2021

Ruhr University Bochum, Bochum, GER

- Lab Supervisor
 - Biophysics
 - Geophysics

Research

- 2026** **Positive Genus Pairs from Amplituhedra**
With Dmitrii Pavlov and Rainer Sinn
[arXiv](#)
- 2025** **Connection Matrices in Macaulay2**
With Paul Görlach, Anna-Laura Sattelberger, Mahrud Sayrafi, Hendrik Schroeder, Nicolas Weiss, Francesca Zaffalon
To appear in JSAG
[arXiv](#)
- 2025** **Taking the Amplituhedron to the limit**
With Rainer Sinn
[arXiv](#)
- 2024** **Uniqueness of MHV Gravity Amplitudes**
With Umut Oktem, S. Paranjape, J. Trnka and B. Zacovic
Le Matematiche - Special Volume on Positive Geometry
[arXiv](#)

AWARDS

- 2023** **International Max-Planck Research School scholarship**
Max-Planck Institute for Mathematics in the Sciences
- 2022** **Excellency Award for Bachelors in Mathematics**
Ruhr University Bochum
- 2017** **UDE Scholarship for Nano-Engineering**
University Duisburg-Essen

ORGANIZED EVENTS

- 2025** **Chow Lectures 2025 by Nima-Arkani Hamed**
MPI MIS Leipzig, [Link](#)
- 2024** **IMPRS COMBO III**
MPI MIS Leipzig, [Link](#)

TALKS & PRESENTATIONS

- 2025** **SIAM: Conference on Applied Algebraic Geometry**
Madison Wisconsin, USA
Taking the amplituhedron to the limit
- 2025** **The Amplituhedron: Structure, Combinatorics, and Positive Geometry**
Les Diablerets, Switzerland
Taking the amplituhedron to the limit
- 2025** **Summer School in Total Positivity and Quantum Field Theory**
Harvard CMSA, USA
Taking the amplituhedron to the limit
- 2025** **The 11th Conference of the Fachgruppe Computeralgebra**
MPI MIS, Germany
Connection matrices in Macaulay2
- 2025** **Geometry Seminar**
University of Oslo, Norway
Taking the amplituhedron to the limit
- 2024** **Special Seminar on Positive Geometry**
MPI MIS, Germany
Uniqueness of MHV gravity scattering amplitudes
- 2024** **Meeting of Scientific Advisory Board**
MPI MIS, Germany
Positive Geometries, [Slides](#)
- 2022** **Seminar on Combinatorics**
Ruhr University Bochum
Combinatorial Models of the Grassmannian, [Link](#)